
Design Document for **questbuddy**

Group **2_jubair_1**

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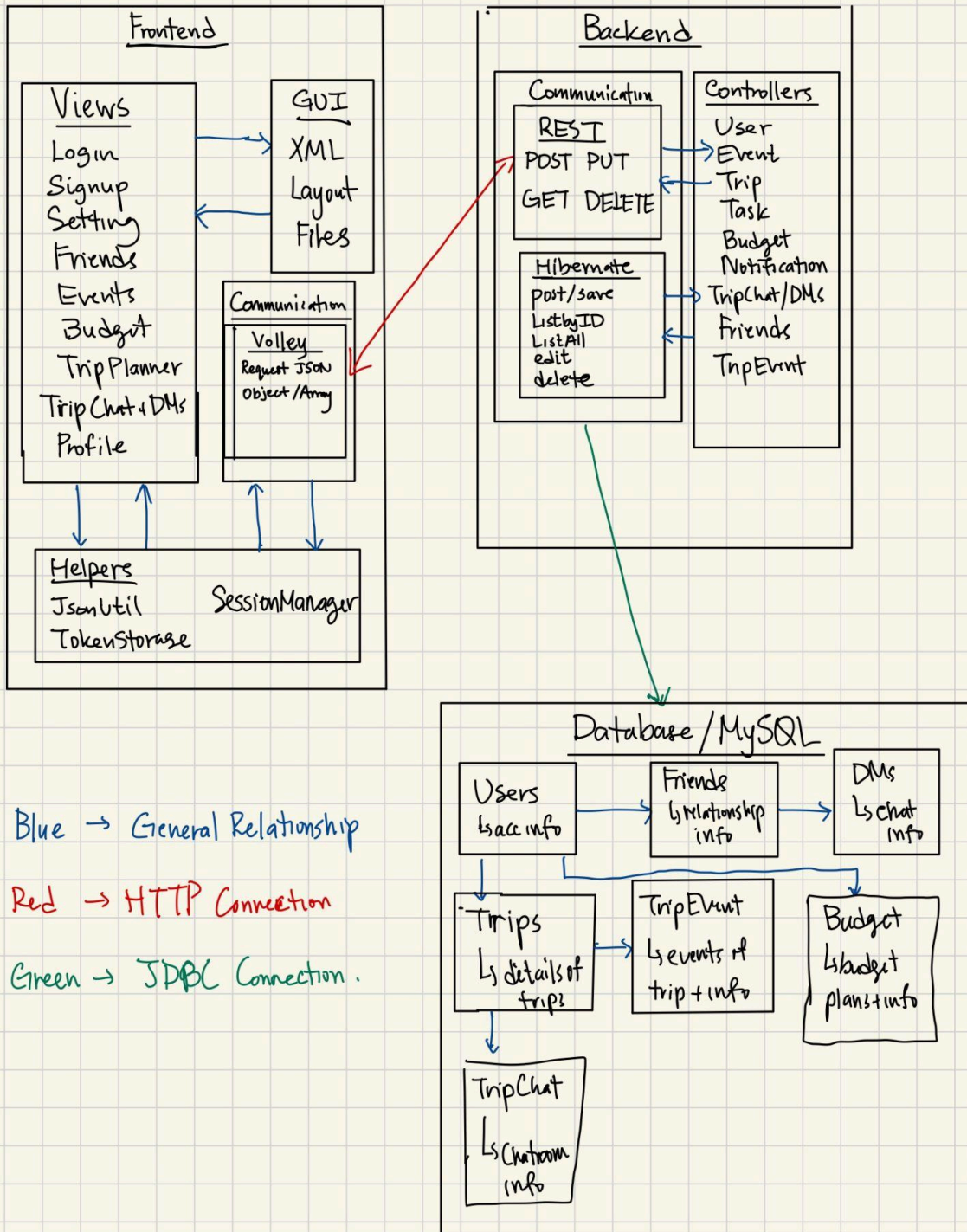
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Block Diagram:

2_jubair_1

Questbuddy.



Block Diagram Description:

Frontend(Android Client):

Logging in and Signing Up

- Login and Signup screens generate pages with the following elements:
 - EditText: Email/username
 - EditText: Password
 - Button: Login / Signup
- When the button is clicked, a POST request is sent with the entered credentials using the Volley communication layer.
- If the backend verifies the user, TokenStorage and SessionManager store session information, and the user is taken to the main app views.

Trips and Trip Planner(User)

- The Trips screen shows the user a list of their trips:
 - RecyclerView: List of trips
 - TextView: Trip name, dates, brief description
 - Button: Create Trip
- Clicking Create Trip sends a POST request with the new trip details to the Trip controller.
- Selecting a trip opens the Trip Planner view, which pulls full trip details (members, basic itinerary, and budget summary) with a GET request.

Budget(User)

- The Budget screen includes:
 - RecyclerView: List of budget items or categories
 - EditText: Amount and description fields
 - Button: Add / Update Item
- When the user adds or edits an item, Volley sends a POST or PUT request to the Budget controller, and the list is refreshed via a GET call.

Friends, Events, and Trip Chat/DMs (User)

- The Friends view displays:
 - RecyclerView: List of friends and pending requests
 - Button: Add Friend / Accept Request

- The Events view shows upcoming trip events retrieved from the backend using GET requests to the Event / TripEvent controllers.
- The Trip Chat/DMs view provides messaging features:
 - RecyclerView: Conversation messages
 - EditText: Message input
 - Button: Send
- Sending a message issues a POST request to the TripChat/DMs controller; message history is loaded with GET requests.

Profile and Settings(User)

- The Profile and Settings screens enable users to view and edit their basic account information and preferences.
- Updates such as name changes or profile details are sent as PUT requests to the User controller.

GUI/Helpers

- All views are defined with XML layout files, which control the UI components and visual structure.
- Helper classes, including JsonUtil, TokenStorage, and SessionManager, handle JSON formatting, token storage, and session tracking.
- The Communication block uses Volley to send JSON Object/Array requests and queue network operations between the Android client and the backend REST API.

Backend(Currently Implemented)

Communication:

- The backend is a Java Spring Boot server exposing a RESTful API.
- Standard HTTP methods are used:
 - POST – Create new users, trips, tasks, budgets, chat messages, friendships, and events.
 - GET – Retrieve users, friends, trip lists, trip details, budgets, chat histories, and event lists.
 - PUT – Update existing entities such as trip info, tasks, or budget items.
 - DELETE – Remove items such as trips, tasks, chats, or friendships when allowed.

Controllers:

- UserController – Manages registration, login, profile viewing, and profile updates.
- EventController / TripEventController – Exposes endpoints for viewing and managing events tied to trips.
- TripController – Handles creating trips, listing trips for a user, and editing trip details.
- TaskController – Manages tasks associated with trips (creation, update, deletion).
- BudgetController – Handles budget plans, limits, and individual budget items.
- NotificationController – Sends and retrieves notifications for trip updates, messages, or invites.
- TripChatDMsController – Manages trip group chats and direct messages between users.
- FriendsController – Handles friend requests, acceptance, and friendship relationships.

Data Management(MySQL Database)

- Data is stored in a MySQL database, accessed via JPA/Hibernate (represented by the green JDBC connection in the diagram).
- Main tables include:
 - Users – Stores basic user account information.
 - Friends – Stores friendship and relationship info between users.
 - DMs – Stores direct message chat data.
 - Trips – Stores details of each trip.
 - TripEvent – Stores events associated with trips.
 - TripChat – Stores messages for trip-based group chats.
 - Budget – Stores budget plans and spending limits for trips.
- The relationships between these tables (indicated by blue arrows in the diagram) enable the backend to link users to their trips, chats, events, and budgets, while maintaining consistency across the entire application.

Database Schema - MySQL Workbench

